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BRIEF REPORT

Clinical Impact of Device-Based Multisensory Monitoring in Heart Failure

A Propensity-Matched Study From the Netherlands

Ugur Aslan, MD,[a] Bert A.C. Zwaenepoel, MD,[a,b] Bas Kirchhof, MD,[a] Bart J.A. Mertens, PhD,[a,c] Michelle Feijen, MD, PhD,[a] Lieselot van Erven, MD, PhD,[a] Laurens F. Tops, MD, PhD,[a] J. Wouter Jukema, MD, PhD,[a,d] Anastasia D. Egorova, MD, PhD,[a,*] Saskia L.M.A. Beeres, MD, PhD[a,*]

Preventing heart failure (HF) hospitalization is crucial for optimizing patient outcomes. Telemonitoring enables early detection of worsening HF, thereby facilitating timely ambulant treatment adjustments and potentially preventing hospitalizations.[1] Many HF patients have guideline-indicated cardiac implantable electronic devices (CIEDs), which provide an opportunity for HF monitoring by continuously tracking individual physiological trends without additional patient effort.

Several CIED manufacturers offer multisensory algorithms that can be activated for HF monitoring. The HeartLogic algorithm (Boston Scientific) and the TriageHF algorithm (Medtronic) continuously monitor multiple physiological parameters and generate an alert when worsening HF is suspected.[2,3]

Although previous studies demonstrated the feasibility and safety of incorporating these algorithms into practice, the impact of these algorithms on clinical outcomes remains unclear. This propensity-matched study investigated the impact of CIED-based multisensory HF monitoring on HF-related hospitalizations.

METHODS

This propensity-matched study included consecutive adults with HF, as defined by the ESC (European Society of Cardiology) guidelines,[4] who had a CIED with active remote monitoring and were under follow-up at LUMC (Leiden University Medical Center) (Leiden, the Netherlands) between May 1, 2023 and February 1, 2025. Patients with complex congenital heart disease, a left ventricular assist device, active malignant disease, and/or renal replacement therapy were excluded.

Patients with a HeartLogic or TriageHF algorithm-compatible CIEDs were followed up according to the CIED-based multisensory algorithm alert-guided care path on top of standard of care (SOC) (**Figure 1A**). Patients with noncompatible devices received SOC and served as control subjects. SOC comprised medical therapy, periodic clinic follow-up, and continuous remote monitoring with 6-monthly device interrogation in accordance with current recommendations.[4,5] Clinical and demographic data were collected from electronic health records; device and transmission data were collected from the device telemonitoring platforms.

Propensity score matching (1:1 ratio, match tolerance of 0.05) was performed using scores that were calculated with logistic regression, in which treatment status was regressed according to baseline characteristics (age, sex, HF origin, NYHA functional classification, mitral regurgitation grade, left ventricular ejection fraction, previous HF hospitalization, and

From the [a]Department of Cardiology, Leiden University Medical Center, Leiden, the Netherlands; [b]Department of Cardiology, AZ Delta, Roeselare, Belgium; [c]Department of Biomedical Data Sciences, Leiden University Medical Center, Leiden, the Netherlands; and the [d]Netherlands Heart Institute, Utrecht, the Netherlands. *These authors contributed equally to this work as last authors.
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ABBREVIATIONS AND ACRONYMS

CIED = cardiac implantable electronic device

ED = emergency department

HF = heart failure

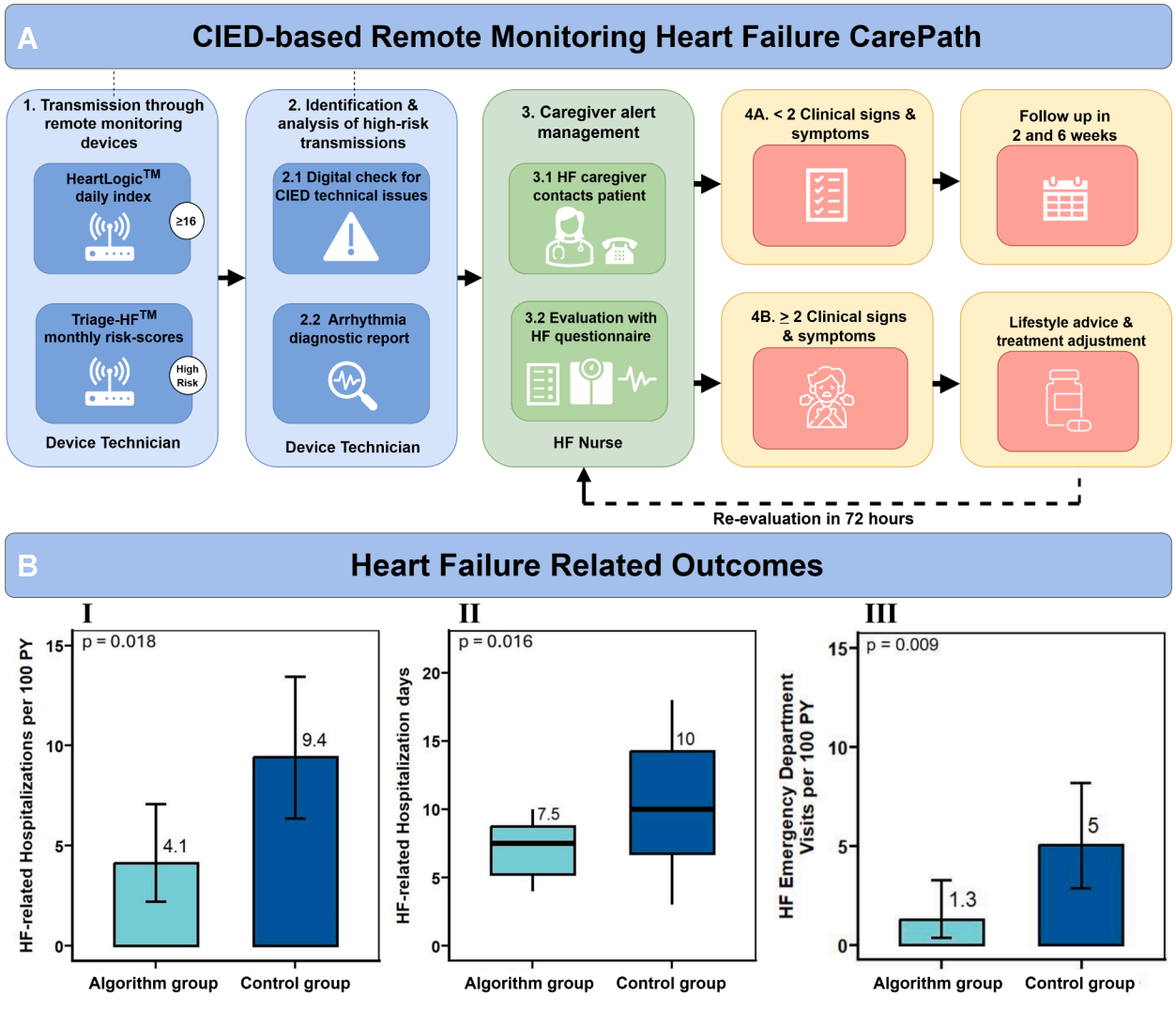
PY = patient-years

SOC = standard of care

cardiac resynchronization therapy). Covariate balance before and after matching was evaluated to ensure comparable groups.

The primary outcome was the rate of HF-related hospitalizations, defined as an unscheduled hospitalization with a primary diagnosis of HF, lasting more than 24 hours or crossing a calendar day. Secondary outcomes included HF admission duration and the rate of HF-related emergency department (ED) visits. An explorative analysis of all cause-mortality was performed. Similarly, the number of alerts, their performance, and the corresponding clinical interventions were evaluated.

FIGURE 1 CIED-Based Remote Monitoring Care Path and HF-Related Outcomes



(A) Device technicians screened alerts for arrhythmias or technical issues and forwarded them to heart failure (HF) nurses, who conducted standardized telephone assessments. Alerts were considered true positive if ≥2 symptoms or signs occurred within 6 weeks. Interventions followed ESC (European Society of Cardiology) HF guidelines, ranging from lifestyle reinforcement to diuretic therapy adjustment or hospitalization. Follow-up was conducted after 72 hours, with treatment escalation as needed. Symptomatic persistent atrial arrhythmias prompted overdrive pacing or cardioversion. The absence of HF symptoms or signs led to re-evaluation at 2 and 6 weeks. If a non-HF-related issue was suspected, patients were referred to their general practitioner. Alerts without symptoms or signs after 6 weeks were classified false positive. (B) (I) HF-related hospitalization rate per 100 patient-years (PY) with 95% CI; (II) median HF-related hospitalization days per hospitalized patient with IQR. (III) HF-related emergency department visit rate per 100 patient-years with 95% CI. CIED = cardiac implantable electronic device.

The study was approved by the LUMC Medical Research Involving Human Subjects Act [WMO] committee division 1 (reference 2024-025).

RESULTS

Propensity score matching resulted in 191 patient pairs. Their median age was 70 years, 76% of these patients were male, 49% had ischemic HF, and the median left ventricular ejection fraction was 39%. In the algorithm group, 110 patients (58%) were monitored with HeartLogic and 81 (42%) were monitored with TriageHF. Follow-up comprised 632 patient-years (PY).

During 314 PY of follow-up in the algorithm group, 226 alerts occurred (mean 0.72 alert/PY) in 94 patients (49%). Of these, 212 alerts were evaluated according to protocol, and 143 (67%) were classified true positive, of which 122 (85%) were primarily HF related and 21 (15%) were not primarily HF related but were clinically actionable. Of the HF-driven alerts, 16 (13%) were managed by lifestyle reinforcement, 80 (66%) were managed by outpatient medication adjustment, 13 (11%) required day admission for cardioversion and/or intravenous diuretic agents, 3 (2%) led to an ED visit, and 10 (8%) resulted in HF hospitalization despite intensified outpatient care.

The HF hospitalization rate was lower in the algorithm group: 4.1/100 PY vs 9.4/100 PY (incidence rate ratio: 0.44 [95% CI: 0.23-0.82]; $P = 0.018$) (**Figure 1B**). Among those hospitalized, median admission duration was shorter: 8 days (Q1-Q3: 5-9 days) vs 10 days (Q1-Q3: 7-14 days) ($P = 0.016$). HF-related ED visits were less frequent in the algorithm group: 1.3/100 PY vs 5.0/100 PY (incidence rate ratio: 0.23 [95% CI: 0.08-0.70]; $P = 0.009$). Death or left ventricular assist device implantation occurred less often in the algorithm group (HR: 0.39 [95% CI: 0.19-0.87]; $P = 0.012$).

The sensitivities of HeartLogic and TriageHF to detect impending fluid retention were 96% and 78%, specificities were 82% and 91%, positive predictive values were 66% and 74%, and negative predictive values were 98% and 92%, respectively.

DISCUSSION

The main findings of this study are that HF patients monitored using the CIED-based multisensory HF-monitoring care path have a significantly lower HF-related hospitalization rate, a shorter admission duration, and a lower rate of ED visits compared with patients receiving SOC.

Most alerts were managed conservatively by means of lifestyle reinforcement and oral diuretic therapy adjustments at home. This finding underscores the feasibility and potential cost-effectiveness of CIED-based monitoring. A few alerts were triggered by primarily not HF-related yet clinically meaningful conditions prompting medical action, thus underscoring the broader clinical utility of algorithm-based monitoring. Shorter hospital stays in the algorithm group likely reflect earlier detection and treatment of congestion that enabled faster recovery when hospitalization was required.

Multisensory CIED-based HF monitoring leverages the already implanted CIEDs and does not require additional invasive procedures while providing a noninvasive and scalable approach. It may therefore provide an alternative to invasive hemodynamic monitoring for selected HF patients with a CIED, if these patients are supported by an effective care path ensuring appropriate responses to alerts.

STUDY LIMITATIONS. The findings of this study should be interpreted in light of its single-center design, with the potential for residual confounding. The generalizability of the study results to other health care systems requires further validation, and the study was not powered to compare different CIED-based algorithms. Upcoming trials, including DANLOGIC-HF (Danish Pragmatic Randomized Trial to Evaluate the Effect of HeartLogic-Guided Management on Heart Failure Outcomes; NCT06099158) and ALLEVIATE-HF (Algorithm Using LINQ Sensors for Evaluation and Treatment of Heart Failure; NCT04452149), will provide additional insight into device-enabled HF monitoring strategies.

CONCLUSIONS

Integrating CIED-based multisensory monitoring within a structured HF care path is associated with lower HF hospitalization rates and shorter hospital stays, as well as fewer HF-related ED visits, compared with SOC. These findings support broader use of CIED-driven remote HF monitoring, and warrant validation in randomized trials.

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ADDRESS FOR CORRESPONDENCE: Dr Anastasia D. Egorova, Department of Cardiology, Leiden University Medical Center, Albinusdreef 2, 2333 ZA Leiden, the Netherlands. E-mail: a.egorova@lumc.nl.

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